

Operator

Orchestrator Graph

LLM Engine

State Checkpointer

1. Submit Intent I_{NL} ('Hamburg to Leipzig...')

2. Forward Translation Prompt (I_{NL}, G_{Sub})

3. Returns S_{PDDL} (AST Validated $v_{struct} = 1$)

4. Reverse Reconstruction (S_{PDDL})

5. Returns I_{recon} + Evaluates Divergence d_{sem}

Semantic Gate: $d_{sem} = 0.42 > \tau_{sem}(0.30)$
Trigger LangGraph interrupt()

6. Atomic Checkpoint: Serialize S_{state} (Thread ID)

7. Present I_{recon} & Request Clarification

Execution Paused (0 tokens)
Operator reviews reconstruction

8. Inject Feedback $\mathcal{F}$ ('Relax hops to 4...')

9. Resume Graph & Reload State ($S_{state}, \mathcal{F}$)

10. Monotonic Update C_{k+1} & Reparse PDDL